Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Baldwin, Kenneth M. Ph.D.		
Project Title:	Force Regulation As a Countermeasure To Muscle Atrophy		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Irvine	**State:**	CA
Zip Code:	92697-4560	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	02/29/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Adams, Gregory (University of California, Irvine) Caiozzo, Vincent (University of California, Irvine) Haddad, Fadia (University of California, Irvine)		
Grant/Contract No.:	NCC 9-58-MA00402		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In recent years considerable attention has focused on identifying exercise paradigms to ameliorate the deficits of atrophy and reduced strength in astronaut muscle properties, especially given that exposure to spaceflight is now minimally six months or longer on the International Space Station (ISS). While there is evidence that exercise of the resistance type is partially successful in reducing the muscle atrophy response, a definitive exercise prescription has not been identified that is suitable for conditioning skeletal muscle given the various constraints of microgravity that impact the homeostasis of astronauts. The primary goals of the proposal are to utilize a rodent resistance training model in order to: 1) ascertain the efficacy of acute bouts of isometric, concentric, eccentric, and concentric-eccentric modes of contraction, as performed under conditions of muscle unloading, in their ability to activate molecular markers indicative of anabolic responses that have been shown to favor a positive protein balance profile in order to blunt the atrophy process; 2) test and optimize the most promising paradigm(s) for its efficacy in ameliorating the atrophy of muscle associated with the model of hindlimb suspension; and 3) determine the efficacy of using resistance training, in combination with other pharmacological interventions (such as protease inhibiters), as to their ability to ameliorate muscle atrophy-induced processes as compared to resistance training alone. Ultimately, the objective is to identify a resistance training prescription that can be efficiently performed (minimizing time constraints), requires simple, yet reliable equipment devices, and is attractive to the astronauts for compliance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project provides insight on practical ways to improve muscle mass and lean body mass in both young and old individuals. It has application to sports training and a means to helping aging individuals maintain stregth in their limb muscles thus reducing accidents and injury.
Task Progress:	In the previous FY, 2004, we demonstrated that there were no clear cut differences in the ability of different contraction modes (isometric actions, concentric-shortening actions and eccentric-lengthening actions) in inducing skeletal muscle hypertrophy via a conventional training paradigm involving a rodent resistance exercise model. This work is summarized in a publication: Adams, G.R., D. C. Cheng, F. Haddad, and K. Baldwin. J. Appl. Physiol. 96: 1613-1618, 2004. This work laid the foundation for performing a series of subsequent studies aimed at ascertaining the role of resistance exercise in ameliorating the atrophy that is caused by limb unloading as induced by the rodent hindlimb suspension model. This is an exquisite model that closely mimics the atrophy that occurs during rodent exposure to space flight. Since isometric resistance exercise protocols represent the simplest mode of resistance exercise and likely a paradigm that reduces the propensity for muscle injury, the study summarized in this report tested the hypothesis that an isometric resistance training (RT) paradigm targeting the medial gastrocnemius (MG) of adult rodents is effective in preventing muscle atrophy during the early stages of hindlimb unloading by maintaining normal activation of the IRS-1/ PI3K/Akt signaling pathway. This pathway has been shown to simultaneously create an anabolic response while inhibiting processes upregulating catabolic processes involving expression of key enzymes in the ubiquitination of proteins for degradation. The findings show that during the five days of unloading 1) absolute MG muscle weight reduction occurred by ~ 20%, but muscle weight corrected to body weight was not different from normal weight bearing controls (P< .05); 2) normalized myofibril fraction concentration and content were decreased; and 3) a robust isometric training program, known to induce a hypertrophy response as outlined above, failed to maintain the myofibril protein content. This response occurred in spite of fully blunting the increases in the mRNA for of atrogin-1, MURF-1, and myostatin, e.g., sensitive gene markers of an activated catabolic state. Analyses of the IRS-1/ PI3K/Akt markers indicated that abundance of insulin receptor subrate-1 (IRS-1) and phosphorylation state of Akt and p70S6 kinase were decreased relative to normal control weight bearing rats, and the RT failed to maintain these signaling markers at normal regulatory level. Our findings suggest that in order to fully prevent muscle atrophy responses affecting the myofibril system during unloading, the volume of mechanical stress must be augmented sufficiently to maintain optimal activity of the IRS-1/ PI3K/Akt pathway to provide an effective anabolic stimulus on the muscle. Studies are being initiated to expand on these important findings in order to generate a better prescription for blunting the atrophy that is observed in this unique rodent model. The work cited herein was accepted for publication in the Journal of Applied Physiology: Haddad, F., G. R. Adams, P. W. Bodell, and K. M. Baldwin. Isometric Resistance Exercise Fails to Counteract Skeletal Muscle Atrophy Processes During the Initial Stages of Unloading. J. Appl. Physiol. (In Press).
Bibliography Type:	Description: (Last Updated: 04/21/2011)
Articles in Peer-reviewed Journals	Adams GR, Cheng DC, Haddad F, Baldwin KM. "Skeletal muscle hypertrophy in response to isometric, lengthening, and shortening training bouts of equivalent duration." J Appl Physiol. 2004 May;96(5):1613-8. PMID: 15075307 , May-2004
Articles in Peer-reviewed Journals	Haddad F, Baldwin KM, Tesch PA. "Pretranslational markers of contractile protein expression in human skeletal muscle: effect of limb unloading plus resistance exercise." J Appl Physiol. 2005 Jan;98(1):46-52. Epub 2004 Aug 6. PMID: 15298986 , Jan-2005